

Explore the data

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv("advertising.csv")

# Quick view of the dataset
print(df.head())
print(df.info())
print(df.describe())

# Check for missing values
print("Missing values:\n", df.isnull().sum())

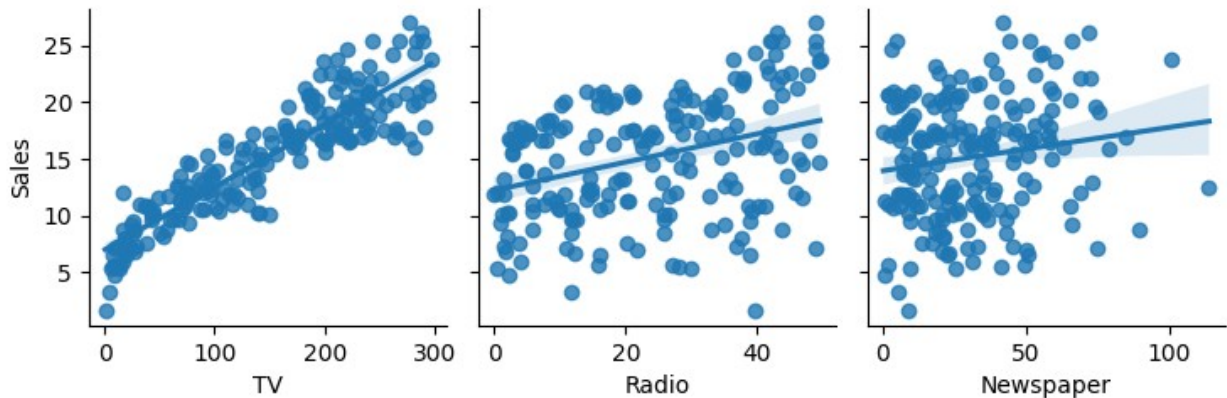
# Visualize relationships
sns.pairplot(df, x_vars=['TV', 'Radio', 'Newspaper'], y_vars='Sales',
kind='reg')
plt.show()
```

```
   TV  Radio  Newspaper  Sales
0  230.1   37.8       69.2   22.1
1   44.5   39.3       45.1   10.4
2   17.2   45.9       69.3   12.0
3  151.5   41.3       58.5   16.5
4  180.8   10.8       58.4   17.9
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0    TV          200 non-null    float64
1   Radio        200 non-null    float64
2  Newspaper    200 non-null    float64
3   Sales        200 non-null    float64
```

```
dtypes: float64(4)
memory usage: 6.4 KB
None
```

	TV	Radio	Newspaper	Sales
count	200.000000	200.000000	200.000000	200.000000
mean	147.042500	23.264000	30.554000	15.130500
std	85.854236	14.846809	21.778621	5.283892
min	0.700000	0.000000	0.300000	1.600000
25%	74.375000	9.975000	12.750000	11.000000
50%	149.750000	22.900000	25.750000	16.000000
75%	218.825000	36.525000	45.100000	19.050000
max	296.400000	49.600000	114.000000	27.000000

```
Missing values:
TV          0
Radio       0
Newspaper   0
Sales       0
dtype: int64
```



Prepare data & train the model

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Define features and target
X = df[['TV', 'Radio', 'Newspaper']]
y = df['Sales']

# Split into training and testing (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

# Train the model
model = LinearRegression()
model.fit(X_train, y_train)

LinearRegression()
```

Make predictions & evaluate the model

```
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np

# Predict sales on test data
y_pred = model.predict(X_test)

# Evaluate the model
mse = mean_squared_error(y_test, y_pred)
```

```
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

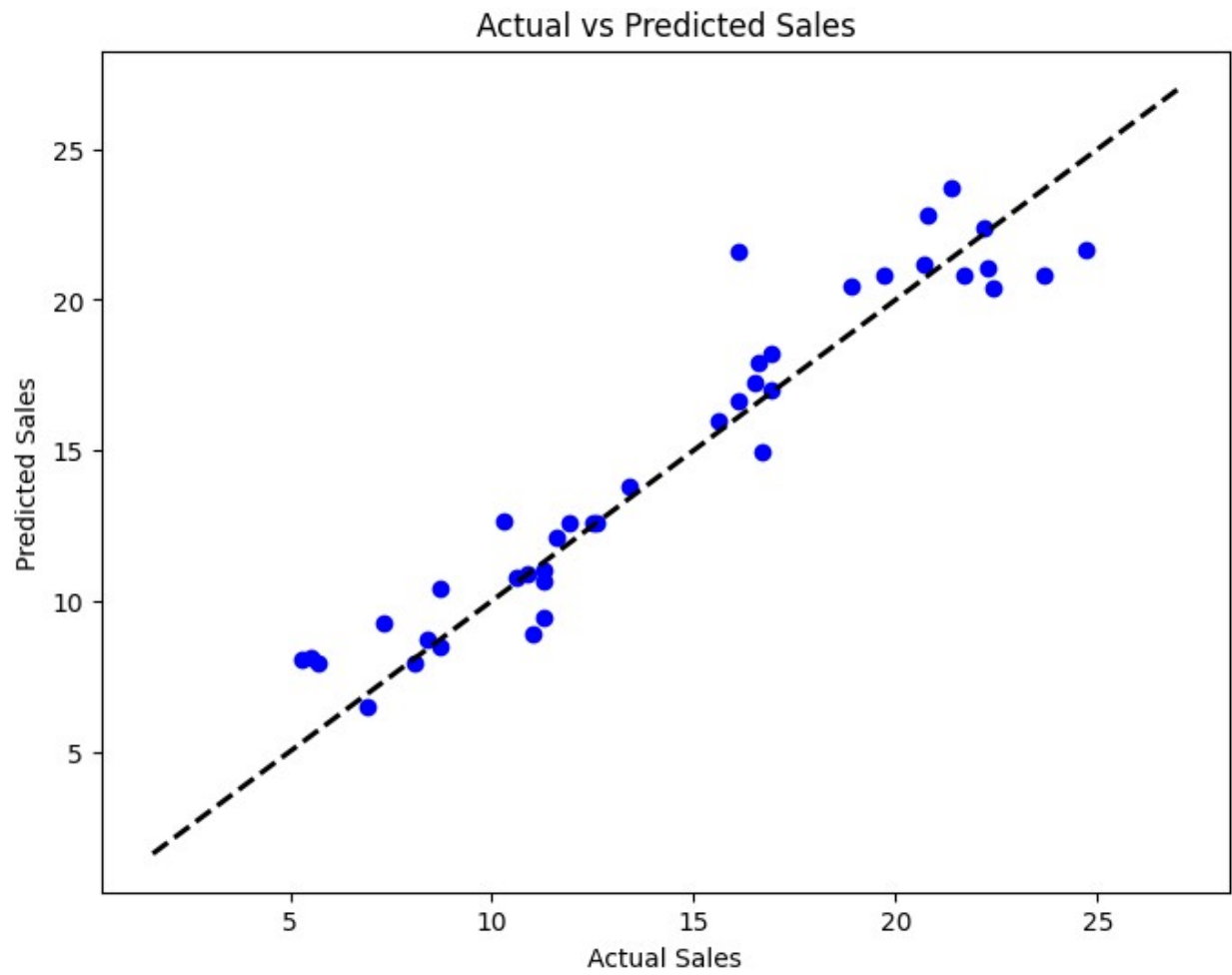
print(f"Mean Squared Error: {mse:.2f}")
print(f"Root Mean Squared Error: {rmse:.2f}")
print(f"R2 Score: {r2:.2f}")

Mean Squared Error: 2.91
Root Mean Squared Error: 1.71
R2 Score: 0.91
```

Visualize predictions & interpret coefficients

```
# Visualize actual vs predicted sales
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, color='blue')
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'k--', lw=2)
plt.xlabel('Actual Sales')
plt.ylabel('Predicted Sales')
plt.title('Actual vs Predicted Sales')
plt.show()

# Check feature importance
coefficients = pd.DataFrame(model.coef_, X.columns,
                             columns=['Coefficient'])
print(coefficients)
```



	Coefficient
TV	0.054509
Radio	0.100945
Newspaper	0.004337